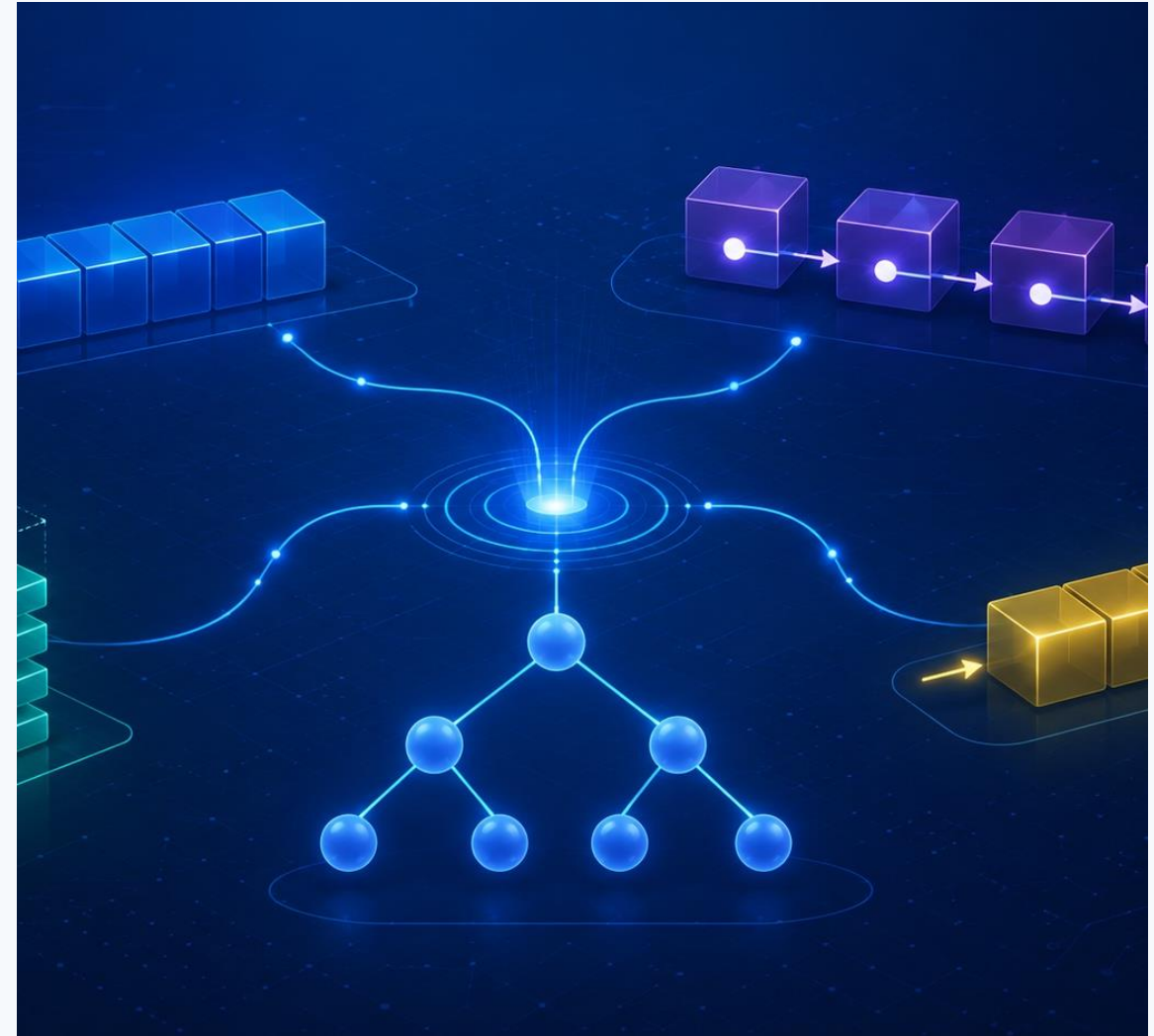


TCS302: Data Structures with C

Unit 1, Lecture 1

Introduction to Data Structures

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Why are we studying Data Structures?

A program is not only a set of instructions. It also needs a smart way to store, access and update data.

Example 1

Searching a contact by name in a phonebook app.

Example 2

Undo operation in an editor or IDE.

Example 3

Finding the shortest route on Google Maps.

Fast access

Less memory waste

Clear logic

Scalable programs

Opening Question: Think Before Coding

Suppose we have marks of 60 students. We need to store them, calculate average, find the highest marks and display all failed students.

1. Store

Where will marks be kept?

2. Access

How will we read each mark?

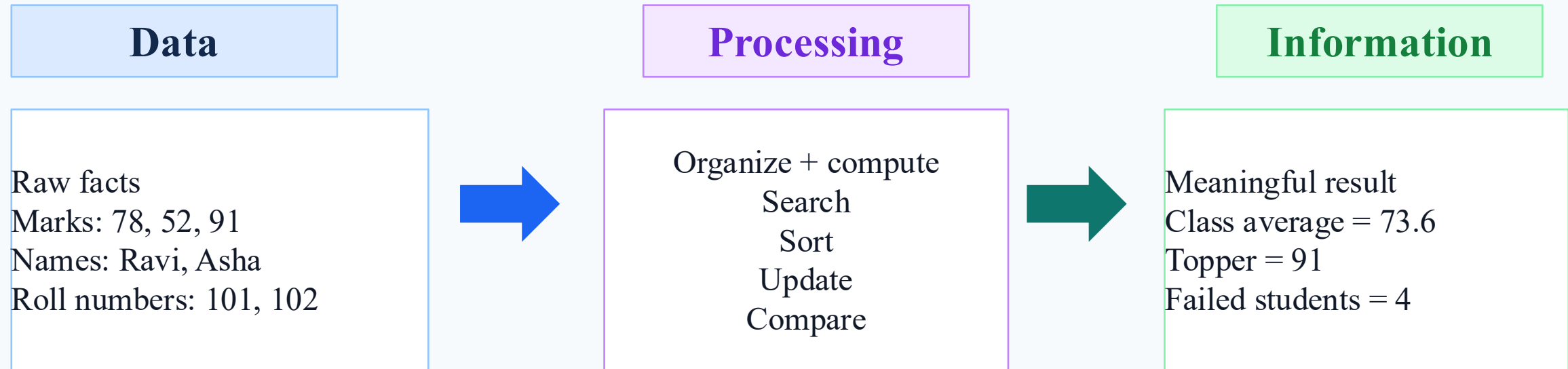
3. Update

How will we change a wrong entry?

4. Process

How will we find average and highest marks?

Data, Information and Organization



Teaching point: Data structures help us move from raw data to useful information efficiently.

What is a Data Structure?

A data structure is a particular way of organizing data in memory so that operations like access, insertion, deletion, searching and traversal can be performed efficiently.

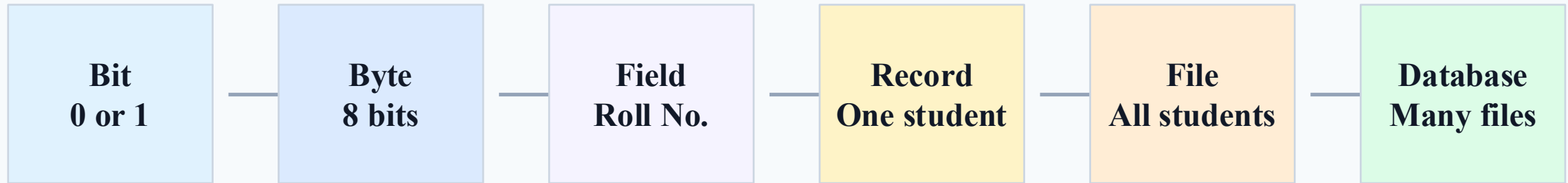
Key idea 1
It stores data.

Key idea 2
**It defines relationships
among data items.**

Key idea 3
**It supports useful
operations.**

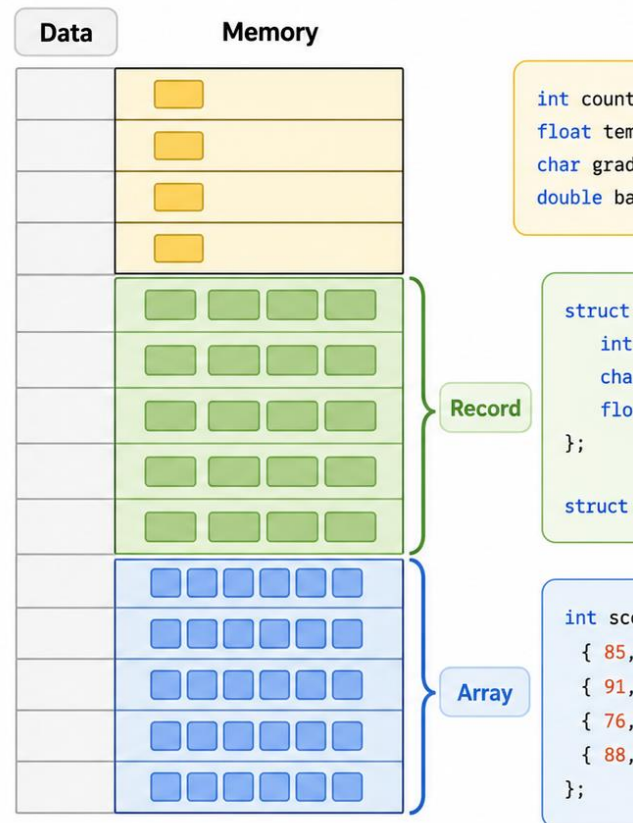
In C, we build data structures using variables, arrays, structures, pointers and dynamic memory allocation.

Elementary Data Organization



Example: A university result system stores bit-level values, characters, fields such as roll number, complete student records, result files and a database.

How Data Looks Inside Memory



When we write a C program, data is stored in memory blocks. The way we group these blocks creates a data structure.

- A variable stores one value.
- An array stores similar values together.
- A structure groups different fields.
- Pointers help create dynamic structures.

Preview: In the next lectures, we will connect C syntax with memory organization.

Common Operations on Data Structures

A data structure is useful only when we can perform operations on it.

Create
Make the structure

Traverse
Visit each item

Search
Find an item

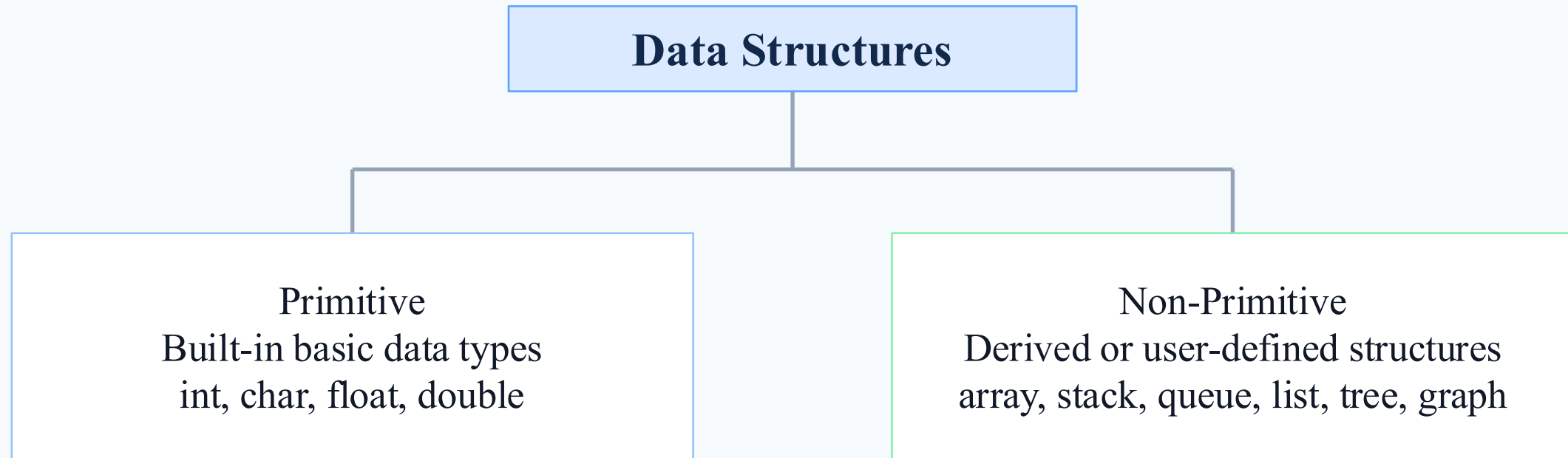
Insert
Add new data

Delete
Remove data

Update
Modify data

Example: In a contact list, adding a new contact is insertion; finding “Satish” is searching; showing all contacts is traversal.

Types of Data Structures: First Classification



In this course, our main focus is non-primitive data structures implemented using C.

Linear and Non-Linear Data Structures

Linear Data Structures

Elements are arranged sequentially. Each item has a clear previous and next item.

Examples

Array, Linked List, Stack, Queue

Non-Linear Data Structures

Elements are arranged hierarchically or as networks. One item may connect to many items.

Examples

Trees, Graphs

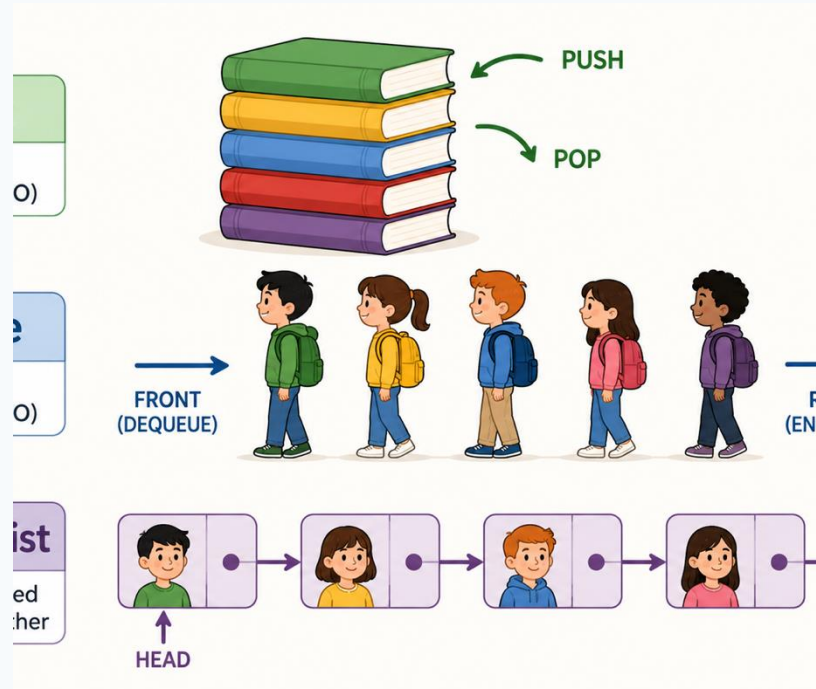
Quick check: Is a railway route map linear or non-linear?

Static and Dynamic Data Structures

Static

Size is fixed before execution or at declaration time.

Example:
`int marks[60];`



Dynamic

Size can grow or shrink during program execution.

Example:
linked list using pointers

Real-World Analogies Make It Easy

Array

Roll-number wise
bench seats
index-based access

Stack

Pile of books
last book added comes
out first

Queue

Line at a ticket
counter
first person enters first

Graph

Cities and roads
many-to-many
connections

Classroom activity: Ask each group to give one more example for any one data structure.

How Do We Choose the Right Data Structure?

**1. What operation is most frequent?
search, insert, delete, traverse**

**3. Is order important?
sequence, priority, hierarchy**

**2. How much data will come?
small, large, growing, fixed**

**4. What memory is available?
fixed storage or dynamic allocation**

**Important rule for students: Never choose a data structure only because it is familiar.
Choose it because it fits the problem.**

Bridge to C Programming

In C, data structures are built step by step from simple language features.

Variables
store single values

Arrays
store similar values

Structures
group different
fields

Pointers
connect memory
locations

```
struct Student {  
    int roll;  
    float marks;  
};  
  
struct Student s[60];
```